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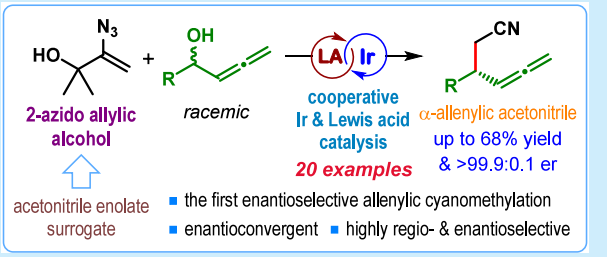
Iridium-Catalyzed Enantioselective Allenylic Cyanomethylation

Aditya Chakrabarty and Santanu Mukherjee*

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ABSTRACT: The poor acidity of acetonitrile restricted its utility as a C2 synthon of a cyanomethyl group for incorporation into organic compounds. We now employed a 2-azido allylic alcohol as an acetonitrile enolate surrogate for the first enantioselective allenylic cyanomethylation. This mild and base-free protocol utilizes cooperative iridium and Lewis acid catalysis for the activation of branched allenylic alcohols to deliver α-allenylic acetonitriles in moderate to good yields with excellent enantioselectivity.



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The ubiquitous occurrence of the cyanomethyl group across a diverse array of bioactive molecules and natural products (Figure 1) inspired the development of methods for

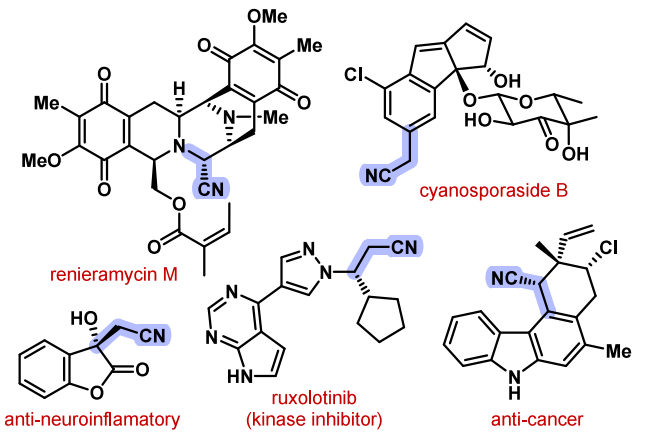


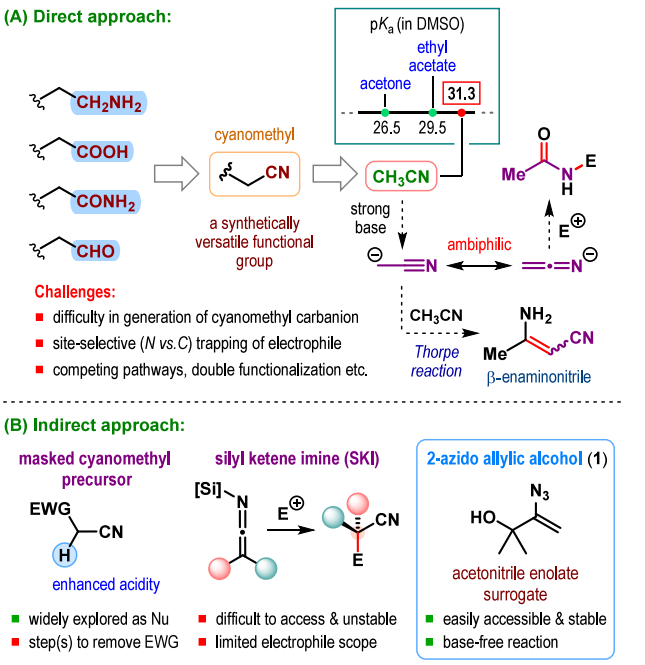
Figure 1. Natural products and bioactive compounds containing the cyanomethyl group.

its introduction to organic compounds.[1] While acetonitrile appears to be the most obvious choice, its use as the precursor of cyanomethyl carbanion comes with serious synthetic hurdles (Scheme 1A).[2]

The attenuated acidity of the α-protons of acetonitrile compared to carbonyls (pK_a of CH_3CN of ~31.3 in DMSO)[3] makes it relatively difficult to deprotonate, and thus, it remains inert toward electrophiles. In fact, the inertness of acetonitrile is a reason for its choice as a solvent in many chemical reactions. Although acetonitrile can be deprotonated using a strong base, the "hardness" and ambiphilic nature of the corresponding cyanomethyl anion makes its selective application as a carbon nucleophile challenging.[4]

Among other pathways, the base-mediated dimerization of acetonitrile to β-enaminonitrile, called the Thorpe reaction

Scheme 1. Direct and Indirect Approaches for the Incorporation of the Cyanomethyl Group



(Scheme 1A),[5] is known to compete with the direct addition of the acetonitrile carbanion to a chosen electrophile. An attempted Pd-catalyzed allylic alkylation of acetonitrile by

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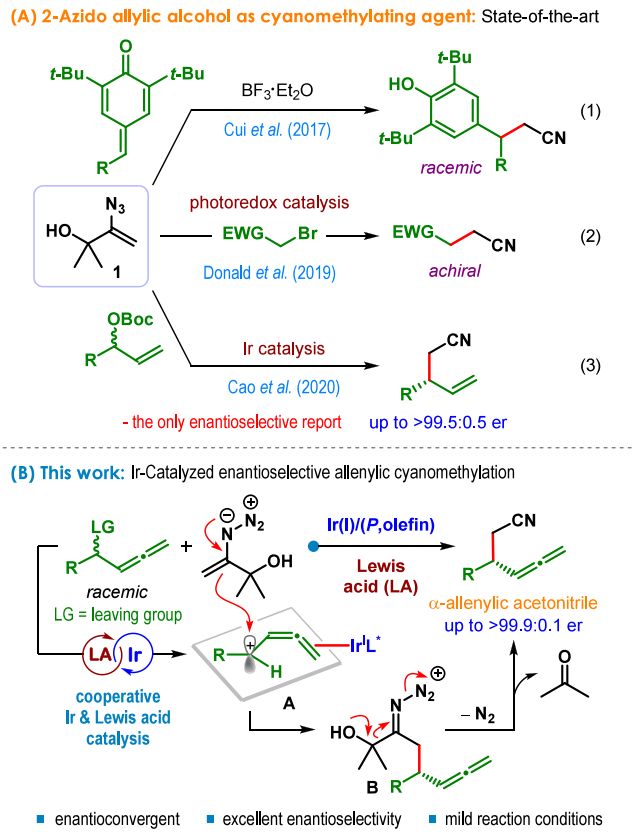
Ding, Hou, and co-workers resulted in α-allylation of β-enaminonitrile through the intervention of the Thorpe reaction.[6] While α-cyanoacetates have been utilized as masked cyanomethyl precursors due to the enhanced acidity of its α-proton, it adds an extra step to remove the carboxylate group in the overall cyanomethylation sequence (Scheme 1B).[7] Despite other notable advancements (e.g., silyl ketene imines as a cyanomethyl carbanion equivalent), only a handful of enantioselective transformations have been reported.[8]

Shibasaki and co-workers developed a general strategy for activating alkyl nitriles through a cooperative combination of Lewis acid with Brønsted base catalysis and accomplished enantioselective addition to a series of electrophiles.[9] In a recent study, Li et al. demonstrated a synergistic Rh and silane catalysis strategy to tame the reactivity of the cyanomethyl anion for enantioselective allylic substitution.[10]

Taking a completely different approach to overcome the challenges associated with the direct use of a cyanomethyl carbanion, Cui et al. in 2017 introduced 2-azido allylic alcohol **1** as an acetonitrile enolate surrogate (Scheme 1B).[11] This stable and easily accessible vinyl azide promises wide applicability due to its apparent compatibility with a variety of catalytic conditions for reactions proceeding under ionic or radical pathways (Scheme 2A, eqs 1 and 2).[11,12] To date, the only application of **1** in an enantioselective transformation is reported by Cao et al. in Ir-catalyzed allylic substitution (Scheme 2A, eq 3).[13]

With our interest in vinyl azides[14] as a surrogate of amide or ketone enolates, we applied them as a nucleophile in Ir-catalyzed enantioselective allylic[15] and allenylic substitution reactions.[16] As a part of this research program, we wondered whether a similar catalytic concept could be extended to 2-azido allylic alcohol (**1**) for achieving enantioselective allenylic cyanomethylation. To the best of our knowledge, no α-allenylation of alkyl nitriles (allenylic cyanoalkylation) has ever been reported, let alone an enantioselective variant. Herein, we present the first catalytic enantioselective allenylic cyanomethylation.[17]

Scheme 2. Cyanomethylation Using 2-Azido Allylic Alcohol as an Acetonitrile Enolate Surrogate

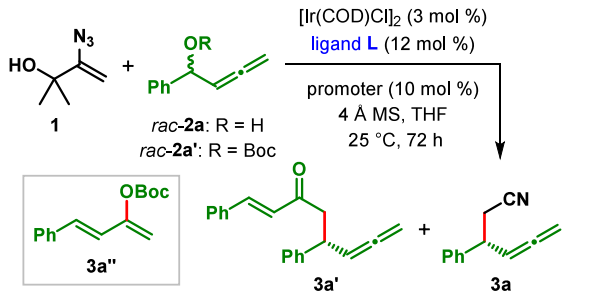


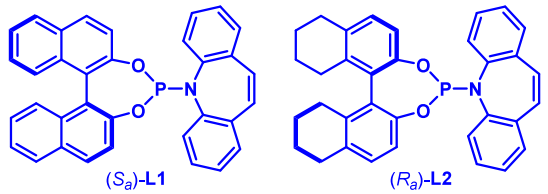
Although the Ir-catalyzed allenylic substitution reaction was first introduced by the Takeuchi group in 2004,[18] its enantioselective variant was not known until the seminal development by Carreira et al. through the application of their Ir(I)/(phosphoramidite,olefin) complex.[19] We envisioned the interception of η^2-Ir(I)-bound allenylic carbocation intermediate **A**, proposed to be generated from a branched allenylic electrophile under cooperative Ir(I) and Lewis acid catalysis,[19,20] with 2-azido allylic alcohol **1** (Scheme 2B). The resulting α-allenylic iminodiazonium intermediate **B** was expected to spontaneously extrude nitrogen and acetone to generate α-allenylic acetonitrile.

To validate our catalytic hypothesis, we began our investigation with the model reaction between 2-azido allylic alcohol **1** and 1-phenyl allenyl alcohol **2a** (Table 1). When the reaction was carried out in THF at 50 °C using 6 mol % of an *in situ* generated Ir complex derived from $[Ir(COD)Cl]_2$ and Carreira's (*P*,olefin) ligand (S_a)-**L1**,[21] in the presence of 10 mol % $Sc(OTf)_3$ as the Lewis acidic promoter and 4 Å MS, the desired α-allenylic acetonitrile **3a** was formed in 34% yield along with the homocoupled product **3a′** as a 1:1 inseparable mixture (entry 1). Despite the low yield, the high enantiomeric ratio (er) of **3a** in this reaction was encouraging and called for further optimization. We have recently disclosed the homocoupling reaction of allenylic alcohols under divergent cooperative catalysis and shown it to proceed through the intermediacy of a cross-dienol intermediate generated via a 1,3-hydroxy transposition of allenylic alcohol.[22] We wondered whether the generation of cross-dienol could be avoided by replacing allenylic alcohol with allenylic carbonate **2a′**. A reaction with 1.5 equiv of **2a′** under similar conditions afforded a slightly improved ratio of **3a** over **3a′** (entry 2). A careful investigation revealed the formation of stable 4-phenyl 1,3-butadien-2-ol-carbonate (**3a″**) via [3,3]-sigmatropic rearrangement of **2a′** in this reaction. Although the use of 2.5 equiv of **2a′** improved the yield of **3a** to 45%, a considerable amount of the homocoupled product was observed (entry 3). When the reaction was carried out at an ambient temperature in the presence of 2 equiv of allenylic alcohol **2a**, α-allenylic acetonitrile **3a** was isolated in 53% yield with 99:1 er and an improved **3a** to **3a′** ratio (5.5:1) (entry 4). Changing the desiccant to 5 Å MS further enhanced the **3a**/**3a′** ratio, even though it was at the cost of the yield (entry 5). Considering the low nucleophilicity of 2-azido allylic alcohol **1** compared to the *in situ* generated cross-dienol, we anticipated that maintaining a low concentration of allenylic alcohols could be beneficial. Accordingly, when **2a** was employed as the limiting substrate, the desired α-allenylic acetonitrile **3a** was obtained almost exclusively (23:1 **3a**/**3a′**) and isolated in 61% yield with 99.8:0.2 er (entry 6). (R_a)-H_8-BINOL-derived (*P*,olefin) **L2** instead of (S_a)-**L1** afforded *ent*-**3a** in a poor yield with reduced er (entry 7), which shows the importance of the BINOL backbone in the (*P*,olefin) ligand for high yield and enantioselectivity. A slightly better reaction profile was

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Table 1. Optimization of Reaction Parameters[a]





entry	1/2	promoter	yield (%) 3a[b]	3a/3a′	er of 3a[c]
1[d]	1:1.5	$Sc(OTf)_3$	34	1:1	99:1
2[d,e]	1:1.5	$Sc(OTf)_3$	30	1.7:1	n.d.
3[d,f]	1:2.5	$Sc(OTf)_3$	45	4.1:1	98.5:1.5
4	1:2	$Sc(OTf)_3$	(53)	5.5:1	99:1
5[g]	1:2	$Sc(OTf)_3$	42	10.5:1	99:1
6	2:1	$Sc(OTf)_3$	(61)	23:1	99.8:0.2
7[h]	2:1	$Sc(OTf)_3$	28	n.d.	5.5:94.5
8[i]	2:1	$Sc(OTf)_3$	(62)	>25:1	99.5:0.5
9[g,i]	2:1	$Sc(OTf)_3$	(65)	>25:1	99.6:0.4
10	2:1	$In(OTf)_3$	35	n.d.	98:2
11	2:1	$Yb(OTf)_3$	30	n.d.	97:3
12	2:1	$Bi(OTf)_3$	21	n.d.	n.d.
13[j]	2:1	$Sc(OTf)_3$	55	5.5:1	99:1
14[k]	2:1	**$Sc(OTf)_3$**	**67**	>25:1	99.5:0.5

[a]Unless stated otherwise, the reactions were carried out on a 0.1 mmol scale using **2a** as the allenylic electrophile. [b]Yields were determined by ^{1}H NMR spectroscopy with mesitylene as the internal standard. Isolated yields after chromatographic purification are shown in parentheses. [c]Enantiomeric ratios (er) were determined by HPLC analysis on a chiral stationary phase. [d]Reaction at 50 °C. [e]Reaction with 1.5 equiv of **2a′**. [f]Reaction with 2.5 equiv of **2a′**. [g]Reaction in the presence of 100 mg of 5 Å MS. [h]Reaction using (R_a)-**L2**. [i]Reaction on a 0.2 mmol scale. [j]Reaction using 20 mol % $Sc(OTf)_3$. [k]Reaction on a 0.4 mmol scale using 400 mg of 5 Å MS.

observed using 5 Å MS as the desiccant when **3a** was formed exclusively in 65% yield with excellent er (entry 9).

Efforts toward enhancing the yield of **3a** further by varying Lewis acidic promoters were ineffective (entries 10−12). Increasing promoter loading to 20 mol % resulted in a poorer **3a**/**3a′** ratio (entry 13). The efficacy of the reaction remained unaltered when the reaction was carried out on a 0.4 mmol scale (entry 9 vs entry 14).

Having optimized the reaction conditions, we set out to showcase the generality of this enantioselective allenylation of 2-azido allylic alcohol **1**. As shown in Table 2, allenylic alcohols bearing electron-rich as well as electron-deficient 1-aryl substituents were well-accommodated, and the corresponding α-allenylic acetonitriles (**3a−3p**) were obtained in moderate to good yields and generally with excellent enantioselectivity. Consistent with the proposed involvement of the allenylic carbocation (**A** in Scheme 2B) in this reaction,[19] allenylic alcohols containing an electron-rich aryl group generally resulted in products with higher yields compared to those bearing electron-deficient aryl substituents. *Para-* or *meta*-substituted aryl rings on allenylic alcohol were found to be better suited in this reaction compared to the *ortho*-substituted ones. In the latter case, the products were obtained in lower yields, although excellent enantioselectivity was maintained. A low yield of **3m** is in agreement with the lesser stability of the corresponding allenylic carbocationic intermediate as well as slow nucleophilic attack of **1**, leading to faster decomposition of allenylic alcohol **2m**. Allenylic alcohols containing a disubstituted electron-rich aryl group (**2k** and **2l**) afforded the products (**3k** and **3l**) in moderate to good yields with high er. Allenylic alcohols **2n−2p** with polyaromatic substituents delivered the products (**3n−3p**) in moderate yield while maintaining outstanding enantioselectivities.

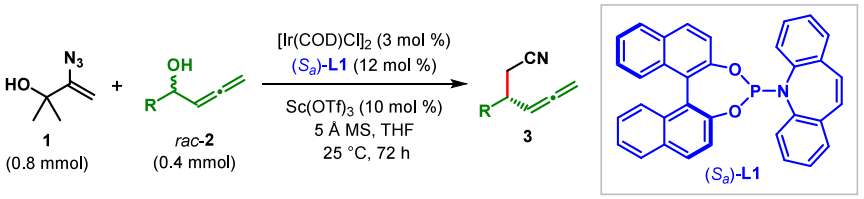
Allenylic alcohols substituted with heterocycles, such as 1,3-benzodioxolane (**2q**) and differently substituted thienyls (**2r** and **2s**), are compatible with our reaction conditions. However, the corresponding products (**3q−3s**) were obtained in relatively low yields. Ferrocene-substituted allenylic alcohol **2t** provided the α-allenylic acetonitrile in only 39% yield with 90:10 er. The lower enantioselectivity points toward the competing background reaction, likely due to the ease of formation and higher stability of the ferrocene-substituted free allenylic carbocation.[23] Aliphatic allenylic alcohols failed to participate in this allenylic cyanomethylation reaction, presumably due to the lower stability of the corresponding allenylic carbocation, which is a limitation of this catalyst system. It is noted that the formation of no other side products could be detected in any of these cases. The low yield of the product in certain cases (**3o−3q**) can be attributed to the decomposition of allenylic alcohol under Lewis acidic conditions.

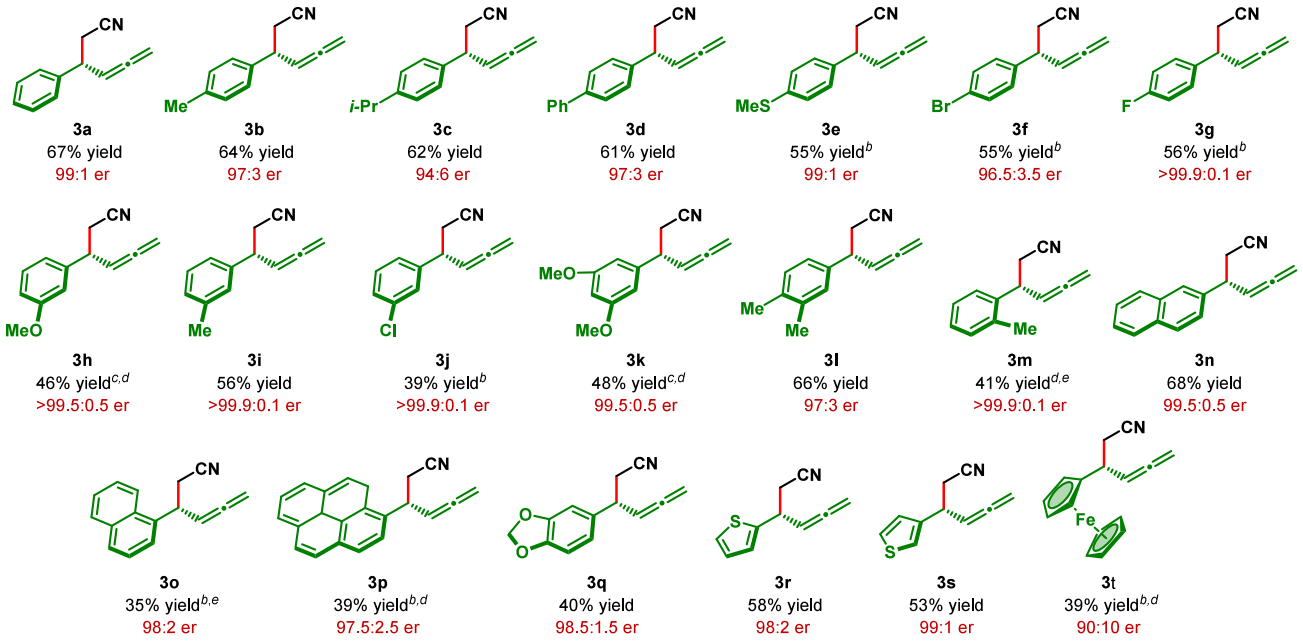
To showcase the scalability of our protocol, a reaction between 2-azido allylic alcohol **1** and allenylic alcohol **2d** was carried out on a 1.0 mmol scale under our optimized reaction conditions (Scheme 3A). The desired product **3d** was isolated in 55% yield with 99.5:0.5 er. The synthetic utility of α-allenylic acetonitriles was illustrated through several synthetic transformations involving **3d** (Scheme 3B). Selective reduction of the cyano group in **3d** with DIBAL-H at 0 °C afforded α-allenylic acetaldehyde **4** in 85% yield. This two-step process represents the first formal α-allenylation of acetaldehyde. The addition of the phenyl Grignard reagent to **4** followed by oxidation of the resulting unpurified secondary alcohol (with 1:1 dr) led to the formation of α-allenylic acetophenone **5** in 91% yield over two steps with 99.5:0.5 er. The absolute configuration of **5** was ascertained to be *R* by comparing its specific rotation to that of a sample obtained through α-allenylation of vinyl azide as a ketone enolate surrogate.[16] The absolute configuration of other α-allenylic acetonitriles shown in Table 2 was tentatively assigned as the same by analogy.

Reductive ozonolysis of **3d** resulted in primary alcohol **6**, which was protected as carbamate **8**. Reduction of **3d** with $LiAlH_4$ and Boc protection afforded *N*-Boc bis-homoallenylic amine **9**, which was subjected to Ag-catalyzed cyclization: 2,3-disubtituted pyrrolidine **10** was obtained as a single diastereomer in 43% yield over three steps with excellent enantioselectivity.[24] No erosion of the enantiopurity was observed in any of these reaction sequences. The Ritter reaction of **3d** with 1-adamantanol (**11**) furnished *N*-adamantyl α-allenylic amide **12** in 67% yield.[25] However, **12**

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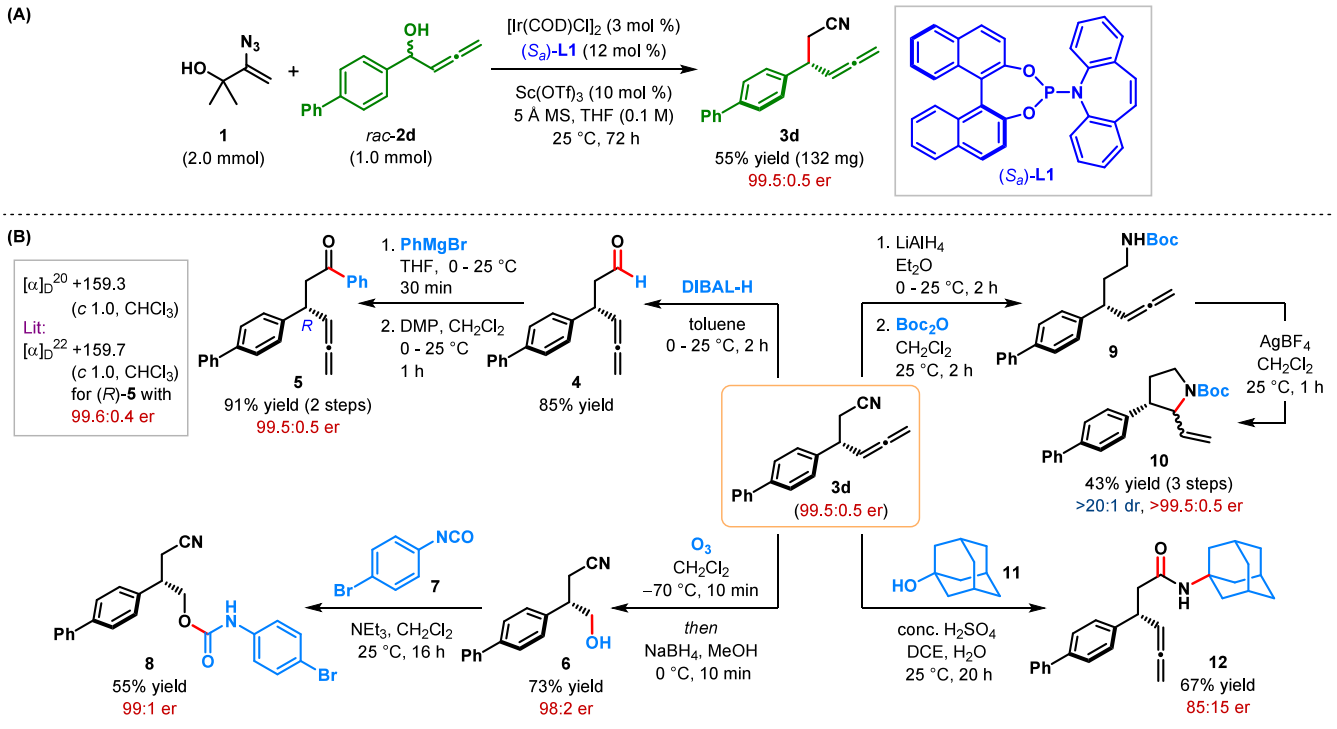
Table 2. Substrate Scope[a]





[a]Unless noted otherwise, the reaction conditions indicated above were followed. Yields refer to the isolated product after chromatographic purification. Enantiomeric ratios (er) were determined by HPLC analysis on a chiral stationary phase. [b]Reaction over 96 h. [c]Reaction using 0.2 mmol of **1**. [d]The reaction was carried out on a 0.2 mmol scale. [e]The reaction was performed at 50 °C.

Scheme 3. (A) Scale-Up Synthesis and (B) Synthetic Elaboration of α-Allenylic Acetonitrile 3d



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was isolated with considerably reduced enantiopurity, likely due to the highly acidic conditions required for this reaction. Nonetheless, this two-step functionalization provides an alternative approach for the enantioselective synthesis of *N*-alkyl-substituted α-allenylic amides.

In summary, we have developed the first catalytic enantioselective allenylic cyanomethylation using 2-azido allylic alcohol as acetonitrile enolate surrogate.[17] This transformation proceeds through the cooperative activation of branched allenylic alcohol by an Ir(I)/(*P*,olefin) complex and Lewis acidic $Sc(OTf)_3$ and delivers α-allenylic acetonitriles in moderate to good yields with outstanding enantioselectivity. Easy accessibility of the substrates, base-free reaction conditions, and ambient temperature are some of the practical advantages of our protocol. Efforts toward the utilization of this acetonitrile enolate surrogate in other catalytic enantioselective transformations are currently underway in our laboratory.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its online Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.5c02686.

Experimental details and characterization data (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Santanu Mukherjee − *Department of Organic Chemistry, Indian Institute of Science, Bangalore 560012, India;* orcid.org/0000-0001-9651-6228; Email: sm@iisc.ac.in

Author

Aditya Chakrabarty − *Department of Organic Chemistry, Indian Institute of Science, Bangalore 560012, India;* orcid.org/0000-0003-3470-069X

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.orglett.5c02686

Notes

The authors declare no competing financial interest.

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